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Patented, Biodegradable Fuel Conditioners and Emulsifiers for the Energy Transition: Enhancing Energy Efficiency and Reducing Emissions to Accelerate Sustainability in Marine Operations and Logistics.

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Abstract

The marine operations and logistics sector faces growing pressure to accelerate progress towards sustainability and decarbonization targets. Despite this, fossil fuels continue to dominate the market due to adoption barriers of alternative energy sources such as hydrogen, ammonia, methanol, and biofuels. Barriers including infrastructure, cost, safety, and compatibility challenges highlight that there is a clear need for transitional solutions that can deliver meaningful improvements within existing fuel and engine systems.

This paper presents a technical evaluation of a patented, organic, biodegradable fuel conditioner designed to reduce fuel consumption, lower emissions, and deliver mechanical benefits across a wide range of marine and land-based applications. The conditioner is compatible with all liquid hydrocarbon fuels and all internal combustion engine types and can be deployed without the need for retrofitting or capital investment. Laboratory tests verified that the conditioner does not alter ISO 8217 fuel specification and contributes to improved lubricity.

A series of independent studies and operational trials were conducted under both controlled and real-world conditions. Fuel consumption reductions ranged from 3% to 7% across marine vessels and rail applications, up to 15% in a controlled generator study, and averaged 8% in land-based logistics. Emissions testing confirmed substantial reductions in greenhouse gases, carbon monoxide (CO), carbon dioxide (CO₂), nitrogen oxides (NO_x), sulfur oxides (SO_x), and particulate matter (up to 96%). Additional operational benefits included improved engine cleanliness, reduced carbon deposits, and lower wear, which were due to the mild detergency properties and the increased lubricity provided by the conditioner's formula.

Ongoing R&D focuses on extending compatibility with next-generation fuels as well as supporting waste oil and slops recycling and onboard emissions monitoring. Compatibility and positive results with several biofuel blends have already been demonstrated, indicating that the solution remains relevant not only for the fossil fuels used presently but also for the evolving fuel mix of the future.

Overall, the findings of evaluations and laboratory testing conducted on the fuel conditioner demonstrate that it is applicable across the full marine logistics supply chain, including onboard vessel engines, port-side generators, heavy-duty equipment, and inland transport vehicles such as trucks and locomotives. This paper highlights that the conditioner offers an immediate technically and commercially feasible transitional

solution delivering efficiency and emissions gains while supporting the sector's broader decarbonization goals and maintaining relevance as the industry transitions to lower-carbon fuels.

Introduction

The International Maritime Organization (IMO) has set ambitious targets for decarbonization of the marine industry. Despite these, fossil fuels will continue to power approximately 95% of shipping in 2030, reducing only to 80% in 2040 and 50% in 2050 (DNV 2023). A full-scale transition away from fossil fuels is still decades away.

Electric propulsion systems using batteries are limited due to their insufficient energy density and long recharging times, particularly for long-distance and deep-sea voyages. Further, alternative fuels like hydrogen, ammonia and methanol face adoption barriers including safety hazards, infrastructure maturity, high training and crew costs as well as significant retrofitting expenses both on vessels and at ports. Biofuels show promise, but there are constraints due to variable quality, stability, cost, limited availability and potential engine compatibility issues (DNV 2023). Therefore, there is a clear need for a transitional solution which can make an immediate impact without the costs, delays and unfeasible infrastructure overhauls.

This paper presents a fuel conditioner with a patented, organic, biodegradable formula. It provides an immediate and scalable solution that requires no capital expenditure, offering a practical, green technology solution to support sustainability and decarbonization across the marine operations and logistics supply chains. The fuel conditioner is applicable to all liquid hydrocarbon fuels, including low-sulfur fuel oil (LSFO), high-sulfur fuel oil (HSFO), marine diesel oil (MDO/MGO), and biofuels. It can be deployed across fleets of all vessel types and engine sizes without the need for capital-intensive retrofitting.

By enhancing fuel efficiency, lowering emissions and extending engine life parts, the conditioner enables maritime operators to optimize fuel performance and meet increasingly stringent environmental regulations. Consequently, enabling the industry to confront transitional challenges of the energy shift by offering a commercially and technically feasible solution while long-term alternatives continue to evolve.

Statement of Theory and Definitions

The patented fuel conditioner stabilizes liquid hydrocarbon fuels and biofuels, optimizes combustion efficiency and completeness therefore reducing emissions and fuel consumption through the combined effects of four key mechanisms.

Fuel Conditioner formula and composition

The conditioner can readily mix with all lighter fuels at ambient temperatures and a range of higher viscosity fuels when the fuel has been heated to working temperatures.

To begin with, when the conditioner is added to the fuel, the surfactants in the formula scavenge small amounts of existing free water in the fuel dispersing it as microscopic droplets forming a stable microemulsion (Ithnin et al. 2018). Primary atomization occurs as the fuel is sprayed into the combustion chamber. During combustion, this microemulsion promotes the secondary atomization effect (Zhou 1992; Fu et al. 2002), further increasing the surface area of the fuel that is available to the air, which in turn improves the combustion profile. Second, the formula donates a small amount of additional oxygen from organic compounds to further aid the completion of combustion, resulting in reduced greenhouse gases and particulate matter production (Armas et al. 2005; Al-Sabagh et al. 2011; Bidita et al. 2012; Vellaiyan and Amirthagadeswaran 2016).

Third, the formula has detergency properties meaning gentle cleaning and decarbonization of engine parts, from the piston heads, filters, injectors, exhaust systems and other components (Meng and Huang 2014). Finally, the formula adds lubricity to fuel which reduces frictional energy losses providing both

fuel efficiencies and engine longevity without capital-intensive changes, an infrastructure overhaul or operational disruptions.

These mechanisms have one holistic goal: to improve the efficiency and completeness of combustion. The conditioner does not directly add cetane points to generate power but instead takes a smarter approach by enabling existing fuels to realize their full power potential by combusting more cleanly and completely. This liberates more power per unit of fuel by converting what would be wasted fuel, in the form of particulate matter, into useful energy that can be harnessed with reductions in harmful emissions and GHGs. Combined with the reduced frictional energy losses from added lubricity and continual decarbonization of the engine parts, this all adds up to noticeable reductions in both fuel consumption and emissions.

The product also offers wider benefits to the engine and subsequent systems resulting in further improvements in fuel efficiency on top of those resulting in improved fuel combustion. A huge amount of energy and efficiency of a system is lost due to friction and wearing of parts. By offering detergency and lubricity enhancements, this waste can be reduced. Decarbonization of pistons crowns, cleaning of fuel and particulate filters (DPF), injector nozzles, turbo and Exhaust Gas Recirculation systems (EGRs), adding lubricity to bore liners are just some of the benefits that all add up to reducing energy wastage and improved fuel economy. These fuel savings vary in degree depending on fuel type, engine type, industry and other factors.

Furthermore, continued use of the product prevents "diesel bug" because the formula prevents water from settling at the bottom of the tank, therefore eliminating the oil-water interface where microbes thrive. Successful testing of diesel samples with up to 1200ppm of water resulted in samples staying bug-free for over 24 months; highlighting how effective the product is for avoiding contamination and growth of microbes.

How does the product reduce emissions and particulate matter?

The combined mechanisms reduce greenhouse gas emissions and formation of particulate matter. The secondary atomization effect is principally the mechanism which lowers peak combustion temperatures which, in turn, reduces the production of thermal NO_x. Particulate matter reductions are noted due to better oxygen availability to the fuel both directly from the formula and increased surface area of the fuel (as a result of secondary atomization) meaning that rather than generating wasteful, hazardous PM emissions, this is converted into useful energy through complete combustion. CO production is reduced as this is also affected by the increased availability of oxygen and is oxidized to CO₂. When the product is first used, often an initial increase in CO₂ emissions occurs. This is a direct result of particulate matter being cleaned and exfoliated from the system, being combusted and oxidized to CO₂. Over time, this small initial elevation in CO₂ output will drop. As fuel economy improves, consumption lowers, overall CO₂ emissions will drop further. SO_x emissions are also reliably reduced in line with consumption ([Lif and Holmberg 2006](#); [Ghojel and Tran 2010](#)).

Test Procedures conducted to confirm safe use of the fuel conditioner

The formula has been independently tested in which it was demonstrated that when the fuel is dosed at the correct ratio (1 part to 2000 parts of fuel) the use of the fuel conditioner does not affect the critical fuel specifications of ISO 8217. Therefore, confirming that the product will not take fuel out of specification and should not be rejected for use by engine manufacturers. Fuel remains within ISO8217 specification with the presence of the conditioner. The testing, conducted by verified regulators using global ASTM standards, was conducted rigorously; both companies tested various blends of distillate and residual fuel before and after dosing with 500ppm of the fuel conditioner. All results provided conclusive evidence that the fuel conditioner does not impact ISO 8217 fuel parameters, confirming full compliance with industry standards.

Independent lubricity testing was also conducted in which 10 distillate fuel samples from various ports were tested in Wear Scar Diameter (HFRR WSD) according to ISO 12156-A. The results showed an average 17.2% reduction using the solution and even brought some fuels which were out of ISO8217 specification based on low lubricity back into acceptable levels as tested.

Research and Development

A dedicated research and development entity has been established to advance forward-thinking solutions focused on next-generation fuels, including novel first and second-generation biofuel blends. Ongoing activities include the development of novel materials intended for use in upcoming formulations of the fuel conditioner. These efforts aim to ensure compatibility with new-generation fuels, enhance the green-credentials of the formula, and support long-term compliance with emerging maritime regulations. As part of this work, further testing will be conducted to evaluate how the conditioner influences the lubricity of these "new-generation" fuels. Further, marketleading innovation is underway focusing on improving waste oil and ship slops recycling processes. Therefore, contributing to supporting more sustainable waste management practices in the marine, industrial and oil and gas sectors, among others.

In addition, further onboard emissions monitoring is scheduled, utilizing class-approved monitoring equipment to evaluate environmental performance under various operating conditions across different voyages.

Description and Application of Equipment and Processes

The fuel conditioner was evaluated for use across a range of internal combustion engines operating on liquid hydrocarbon fuels, with the objective of assessing its potential impact within the marine operations and logistics sector. The product has been assessed for applicability throughout the entire logistics cycle, from ship main and auxiliary engines to land-based equipment such as generators, trucks, locomotives, tampers and heavy-duty equipment utilized on ports.

Multiple independent studies and operational trials were conducted to assess the conditioner's potential for reducing fuel consumption and emissions, and to evaluate its impact on engine cleanliness. In each trial, a structured and specific evaluation protocol was implemented by equipment owners. Testing protocols began with the collection of baseline data on fuel consumption under standard operating conditions, over a defined pre-treatment period. Following this, the fuel conditioner was introduced into the fuel system. Dosing did not require any specialized equipment or procedures; the additive was introduced directly into the fuel tank, typically prior to fuel bunkering, as onboard fuel agitation facilitates dispersion. Post-treatment performance data was then gathered over a corresponding operational timeframe and analyzed.

Emissions testing was undertaken by independent, third-party bodies with globally recognized credentials. Studies were conducted on a diesel generator, locomotives and tampers in which stack emissions and particulate matter were monitored. Standard testing methods were applied to evaluate changes in levels of emissions and particulate matter before and after treatment of the fuel with the conditioner.

Presentation of Data and Results

A series of independent studies and field evaluations were conducted across multiple internal combustion engines and fuel types to assess the effects of the fuel conditioner under both real-world and controlled conditions. The collected data indicated observable improvements in fuel efficiency, emissions profiles, and operational parameters.

Across the marine applications, fuel savings ranged from c.3% to 7%, depending on vessel type and operating conditions. Land-based trials showed an average reduction of 8%, while a dedicated generator study using EN590 diesel recorded fuel savings of up to 15%.

Emissions testing consistently demonstrated reduction in particulate matter (PM10 and PM2.5) by 90-95%, along with lower concentrations of greenhouse gases over both short-term and extended operating periods. Operational observations also noted improvements in engine lubricity, reduced formation of carbon deposits and lacquer on internal components, and a decrease in fuel system sludge accumulation.

The following sections present key data from selected evaluations.

Comprehensive evaluation of the conditioner on commercial vessels across different vessel types and fuel types over a thirteen-month period

An extensive thirteen-month evaluation of the fuel conditioner was conducted by a commercial vessel operator. The study monitored fuel consumption and engine performance across a diverse fleet of tankers, bulk carriers, vehicle carriers, and chartered container ships. These vessels operated on a range of marine fuels, including Heavy Sulfur Fuel Oil (HSFO), Very Low Sulfur Fuel Oil (VLSFO), and B30 biofuel blends.

The analysis was based on time-normalized fuel consumption, using 24-hour operational periods compared against fuel usage measured in metric tons (MT). Vessel speed was excluded from the evaluation, although it can directly affect the fuel consumption. Additional filters were applied to eliminate data inconsistencies and improve reliability: data points were excluded if they contained missing calendar entries, incomplete main engine load readings, fuel types outside the study scope (e.g., Marine Gas Oil), or if the conditioner was not in use during the period. The results are based only on entries for periods of 24 hours of sailing. Main engine load ranges were allowed to vary within controlled limits to maximize the number of comparable data points per vessel.

Furthermore, some vessels alternated between B30 and HSFO frequently depending on sailing routes and port availability. It was noted that lower-viscosity fuels such as B30 and VLSFO would likely offer improved fuel savings due to better mixing characteristics. During the evaluation period, the conditioner was dosed into the fuel system after drip sample validation was conducted rather than directly into the receiving tank. It was anticipated that dosing at the receiving tank would have promoted more effective blending and yielded further fuel consumption reductions.

Fig. 2 summarizes the average fuel savings observed during the evaluation period. Each bar represents a distinct vessel, except for the Vehicle carrier category, which reflects the average results from two vehicle carriers. The results indicate fuel savings ranging from 3.04% to 4.19%, depending on vessel type and operating conditions.

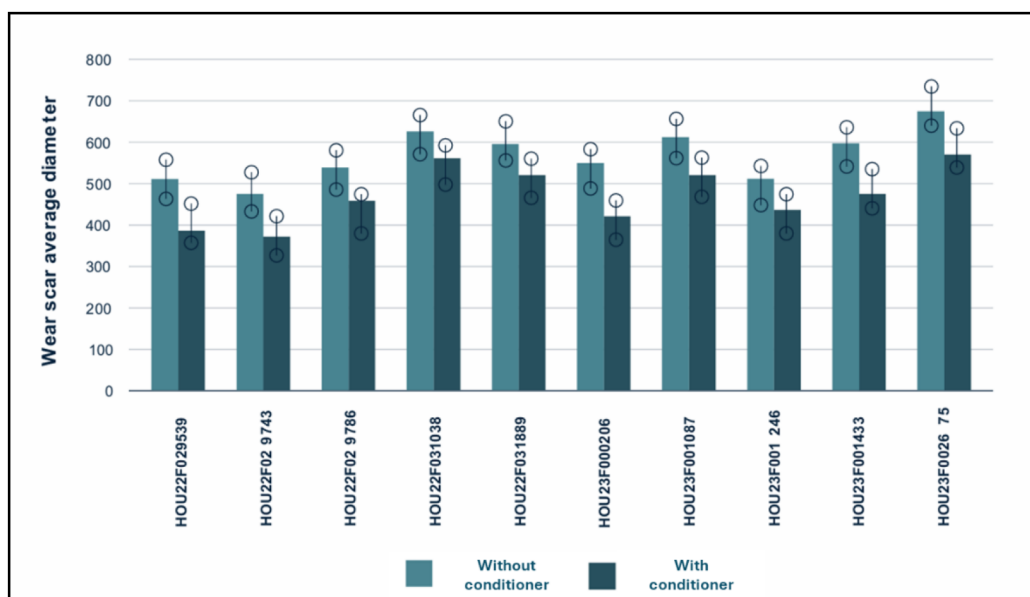


Figure 1—Results of scar-testing, given Dosing Rates 500ppm - Wear Scar Diameter of sample with and without fuel conditioner

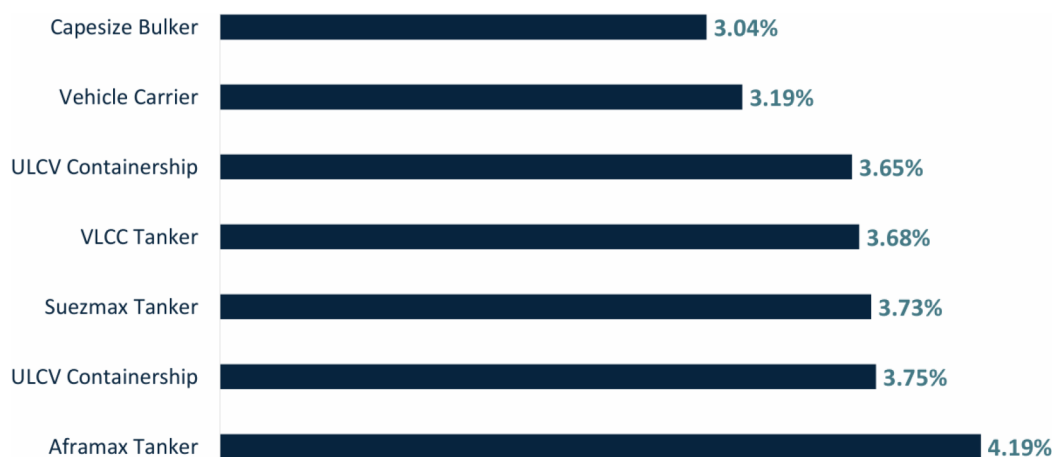


Figure 2—Fuel Savings measured across the vessel types during the thirteen-month evaluation of the fuel conditioner with a commercial vessel operator

Notably, some vessels operated with biofuel blends, including B30, during the assessment period. No adverse effects on fuel performance or engine operation were observed in these cases, suggesting compatibility between the fuel conditioner and bio-derived fuels.

Specific Fuel Oil Consumption (SFOC) was monitored on one vessel using smart ship sensor data. As shown in Fig. 3 and Fig. 4, the use of the fuel conditioner results in measurable reduction in both overall consumption and variability. Post-treatment SFOC values were lower and more consistent, reflecting enhanced fuel efficiency across operations.

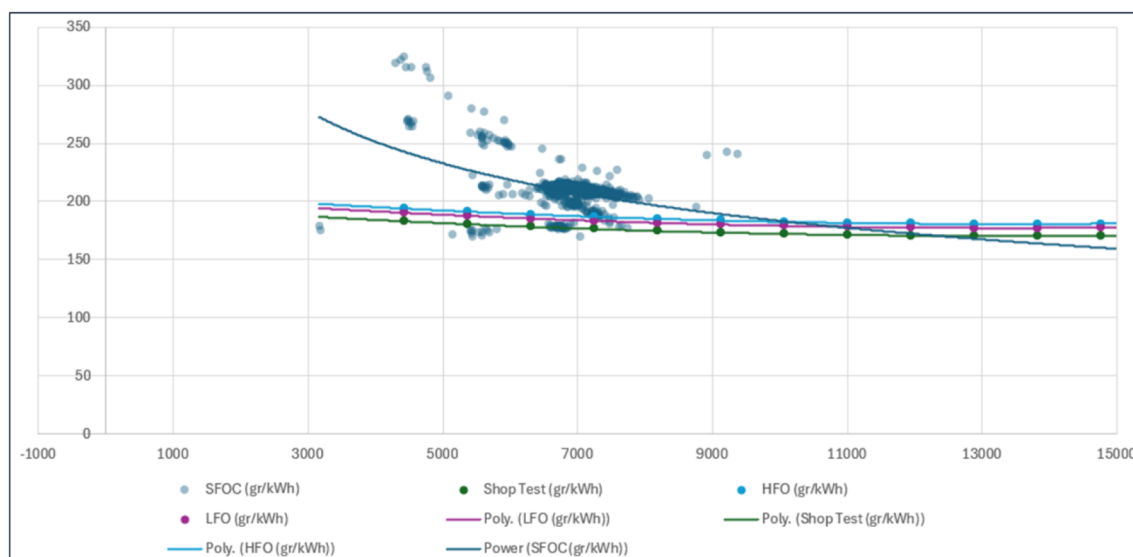


Figure 3—SFOC measurements before conditioning the fuel on one vessel during the evaluation of the fuel conditioner with a commercial vessel operator

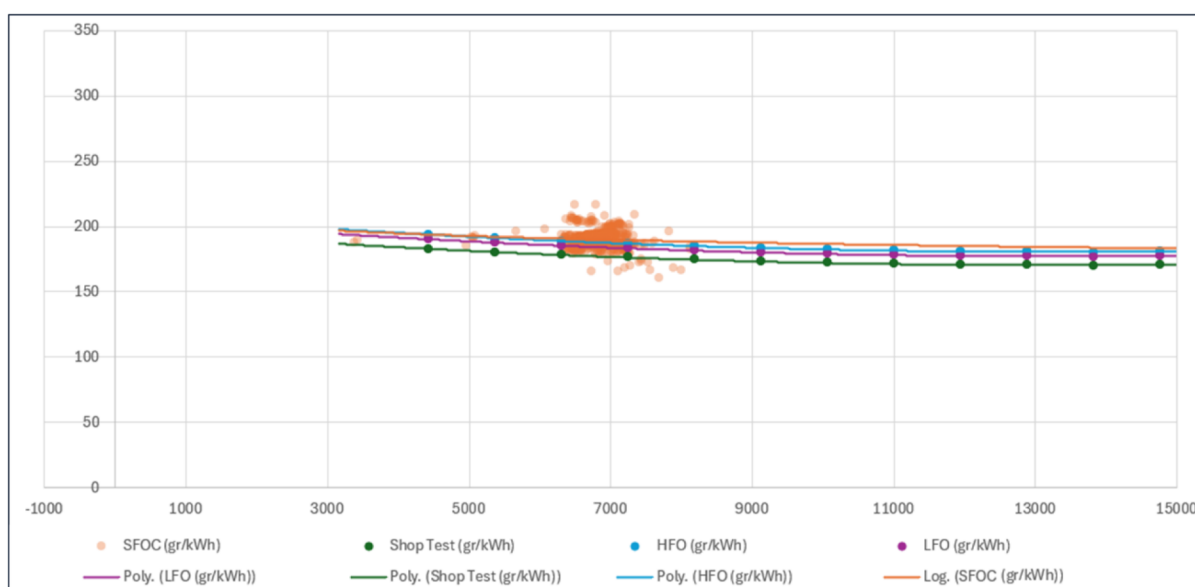


Figure 4—SFOC measurements after conditioning the fuel on one vessel during the evaluation of the fuel conditioner with a commercial vessel operator

The evaluation period revealed additional operational benefits related to engine cleanliness and maintenance. Onboard inspections conducted during and after the trial noted reduced carbon deposits, significant reductions in black smoke generation, and visibly cleaner combustion components. These improvements reduced strain on critical engine parts, contributing to smoother engine operation and fewer maintenance issues. Vessel personnel, including captains and chief engineers, reported improved fuel system cleanliness, effective operations of fuel purifiers with minimal sludge accumulation, absence of carbon accumulation on piston crowns and ring lands, cleaner piston rings and grooves, and fuel injectors free from sludge formation. Photographic evidence (Fig. 5) illustrates the reduction in lacquering along cylinder lining walls, while Fig. 6 depicts visibly cleaner piston crowns and ring lands following extended use of the conditioned fuel.

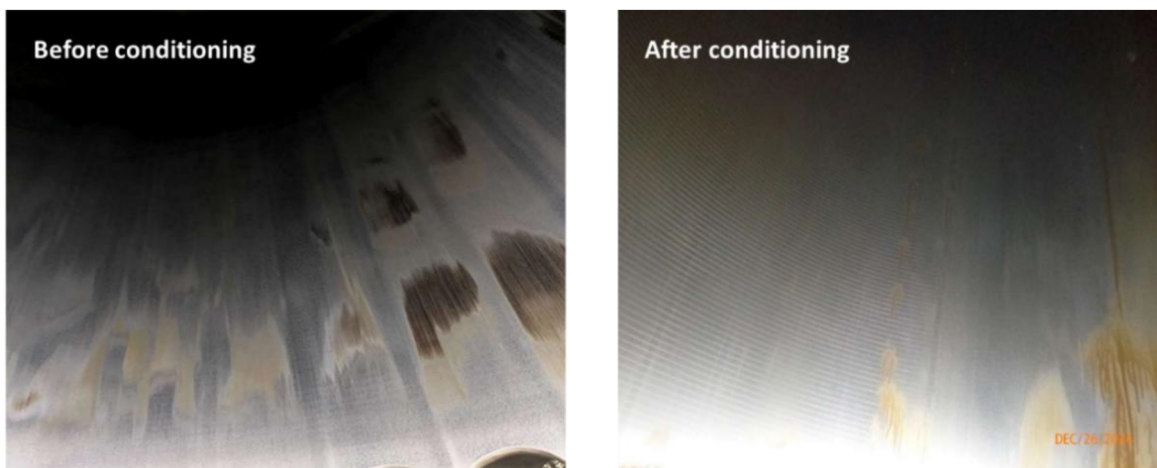


Figure 5—Cleared lining walls witnessed on a vessel during the evaluation of the fuel conditioner with a commercial vessel operator

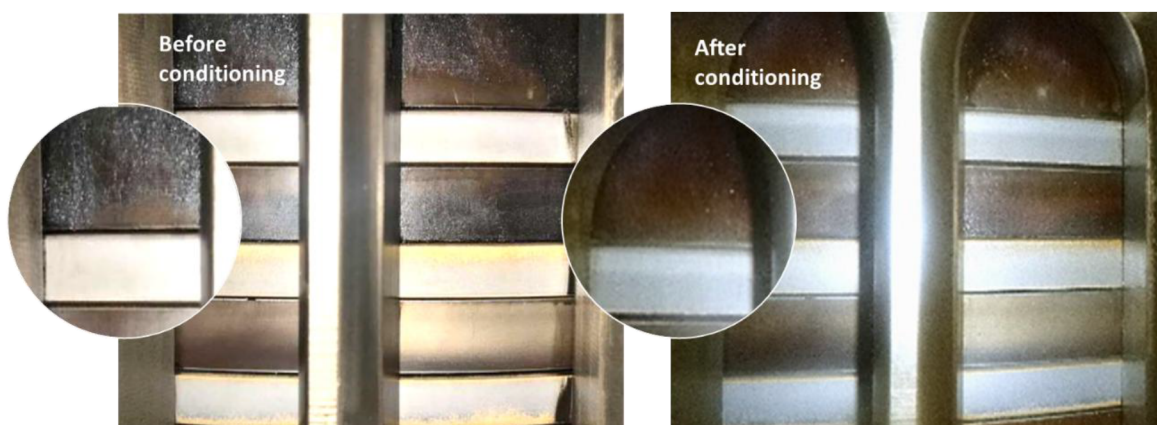


Figure 6—Cleaner crown and ring lands witnessed on a vessel during the evaluation of the fuel conditioner with a commercial vessel operator

Sludge generation was quantitatively measured on one vessel. Baseline data were recorded over 82 days of operation using untreated fuel, followed by an 83-day period using the fuel conditioner. As summarized in [Table 1](#), total sludge generation was reduced by 12.5%, and daily sludge output declined by 13.6%. It is important to note that the conditioner does not primarily treat sludge, but as shown by these results, it can help prevent further generation of sludge.

Table 1—Sludge generation data measured on one vessel during the evaluation of the fuel conditioner with a commercial vessel operator

	No. of Days	Sludge Generated (M ³)	Avg. Per Day (M ³)	Scavenge Drain Box (M ³)
Without Fuel Conditioner				
	6	6.1	1.01	1.1
	14	5.7	0.4	1
	25	29.5	1.18	1.9
	3	5.2	1.7	0.1
	34	25.4	0.74	3.5
Total	82	71.9	0.88	7.6
With Fuel Conditioner				
	12	12.7	1.06	0.8
	15	8.4	0.56	1.8
	26	20.2	0.78	2.5
	30	21.6	0.72	2.8
Total	83	62.9	0.76	7.9

Exhaust gas temperatures were also monitored as an indicator of combustion efficiency. A consistent decrease in exhaust temperatures was recorded after the introduction of the fuel conditioner, as shown in Table 2. This trend remained evident even when accounting for variations in engine load. High in-cylinder temperatures are a known contributor to the production of Thermal NO_x formation. The reduction in exhaust gas temperatures indicated more complete combustion and supports the likelihood that the conditioner effectively reduced Thermal NO_x production resulting in lower measured emissions.

Table 2—Exhaust gas temperature data measured on one vessel during the evaluation of the fuel conditioner with a commercial vessel operator

	Without Fuel Conditioner					With Fuel Conditioner				
Month (Load, MCR%)	Jan (68%)	Feb (68%)	Mar (65%)	Apr (66%)	May (45%)	Jul (60%)	Aug (43%)	Oct (54%)	Nov (44%)	Dec (47%)
Reading 1	319	346	351	341	345	335	328	304	323	321
Reading 2	302	311	328	314	325	320	313	287	304	307
Reading 3	298	307	325	307	312	306	294	274	288	289
Reading 4	322	328	346	331	329	328	315	303	311	317
Reading 5	301	321	327	342	313	329	300	293	295	298
Reading 6	302	314	327	322	312	328	302	301	289	296
Reading 7	301	294	315	300	297	311	288	271	274	285
Reading 8	303	321	328	322	319	331	307	299	303	308
Reading 9	312	325	342	331	319	330	314	298	301	307
Reading 10	288	304	318	302	297	309	284	270	285	286
Reading 11	338	340	356	343	341	352	328	313	322	321
Average	307.82	319.18	333	323.18	319	325.36	306.64	292.09	299.55	303.18

Following the consistent results across multiple vessel types, the shipping operator elected to implement the fuel conditioner across a larger portion of its fleet and continue monitoring its effectiveness. This decision was driven by the potential for the observed benefits to contribute to environmental targets, such as compliance with port air quality regulations, and to operational objectives, including reduced carbon allowance liabilities under the Emissions Trading Systems and improved Carbon Intensity Indicator (CII) ratings for vessels.

Further work was conducted by the operator to independently test the impact of the conditioner on the lubricity on marine fuel. A High-Frequency Reciprocating Rig (HFRR) test on Low Sulfur Marine Gas Oil (LSMGO) fuel showed a 21.5% average lubricity improvement.

The next phase of evaluation by the shipping operator will involve extending the study to include additional fuel types, notably Low-Sulfur Marine Gas Oil (LSMGO) and higher bio-content fuels such as B100 biofuel blends, to assess the conditioner's performance under varying combustion characteristics and fuel viscosities. Further work will focus on long-term tracking of engine wear, injector cleanliness, and fuel system deposits over extended duty cycles. Onboard carbon dioxide monitoring is also planned to evaluate the product's impact in real-time operations across a range of voyage conditions. Collectively, these efforts aim to strengthen the dataset, explore fuel-conditioning effects in more diverse operating conditions, and contribute to industry knowledge on low-emission, high-efficiency marine fuel strategies.

Overall, the results of this evaluation support the technical effectiveness of the fuel conditioner across a range of vessels and fuel types while aligning with the broader sustainability objectives for the shipping company and the marine sector.

Independent study analyzing fuel consumption and emissions on a land generator

An independent third-party study was conducted using a 100 kVA soundproof diesel generator operated under controlled conditions to minimize environmental variability typically encountered in marine and transport applications. The fuel was dosed with the conditioner and the fuel consumption and exhaust emissions from the generator were monitored over a five-week period.

The generator was operated at a constant 80% load using EN590-compliant diesel fuel. After one week of operation with the treated fuel, fuel consumption was measured as 12.9 L/h. This reduced to 11.06 L/h after a further two weeks, corresponding to a 14.5% reduction. As shown in [Fig. 7](#), this lower consumption was sustained over the subsequent two weeks, with an average of 11.22 L/h recorded.

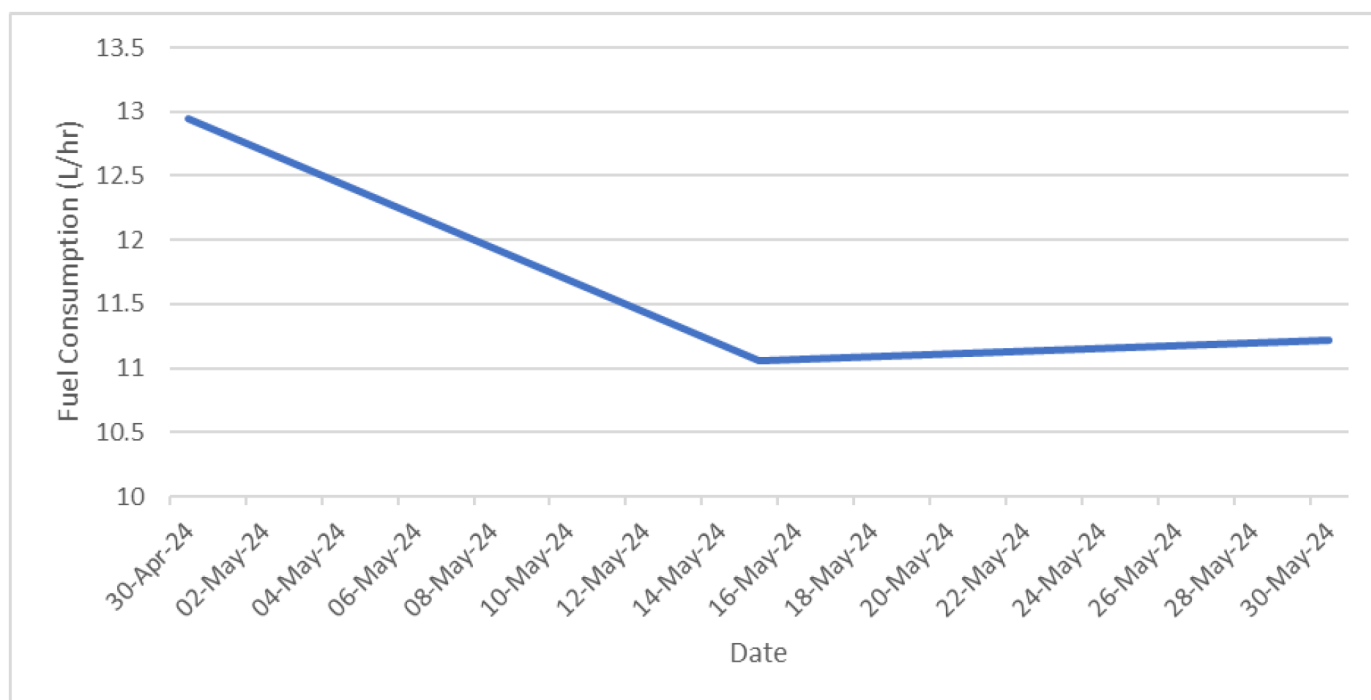


Figure 7—Average fuel consumption in Litres per Hour recorded during the generator study

Emissions were measured at three intervals: pre-treatment, one hour after dosing, and after five weeks of continuous operation. Results, summarized in Table 3, demonstrated substantial reductions in particulate matter emissions, with PM₁₀ and PM_{2.5} levels decreasing by over 96%. SO_x emissions reduced by approximately 64%, while NO_x and CO reduced by approximately 14% and 6.5%, respectively. CO₂ emissions showed a net reduction of 26.7% over five weeks, despite a minor increase observed at the one-hour mark. This initial increase is down to the combustion of pre-existing carbon residues within the engine system, consistent with a shift toward more complete combustion.

Table 3—Emissions and particulate matter results recorded during the generator study

Parameter	Units	Baseline	1 hour	Diff (%)	5 weeks	Diff (%)
PM ₁₀	mg/m ³	7.59	3.05	-59.8	0.29	-96.2
PM ₁₀ Emission Rate	g/h	1.66	0.67	-59.6	0.06	-96.4
PM _{2.5}	mg/m ³	7.44	2.94	-60.5	0.17	-97.7
PM _{2.5} Emission Rate	g/h	1.63	0.64	-60.7	0.04	-97.5
SO _x	mg/m ³	50	2.29	-95.4	18	-64
SO _x Emission Rate	g/h	3.16	0.14	-95.6	1.16	-63.3
NO _x	mg/m ³	456	437	-4.2	390	-14.5
NO _x Emission Rate	g/h	100	96	-4	86	-14
CO	mg/m ³	140	120	-14.3	133	-5
CO Emission Rate	g/h	31	26	-16.1	29	-6.5
CO ₂	%v/v	5.59	5.84	4.5	4.1	-26.7

These findings demonstrate the applicability of the fuel conditioner in marine logistics and port operations, particularly for shore-based generators and auxiliary systems, contributing to cleaner port environments and improved compliance with local air quality regulations. The significant reductions in

fuel consumption and emissions observed under controlled conditions are directly relevant to the maritime vessels. Given the observed combustion improvements and emissions and particulate matter reductions achieved, it is reasonable to anticipate comparable benefits when the fuel conditioner is applied onboard vessels, supporting broader efforts to reduce environmental impact across marine operations.

Other Marine Evaluations

In addition to detailed trials on commercial vessels and generators, the fuel conditioner was evaluated across a range of marine vessels including cruise vessels, river cruise ships and private yachts.

An eight-month evaluation was conducted on a 51,068 gross tonnage cruise vessel powered by six Wartsila-Sulzer diesel engines operating on VLSFO and Marine Gas Oil (MGO). The study recorded an average fuel saving of 3.4%, alongside a visible reduction in black smoke emissions from the vessel's stack. Additional onboard observations included reduced soot accumulation on external surfaces and decks, improved air quality, and reduced maintenance demands. These results reflect a reduction in both operational fuel costs and environmental impact, supporting compliance with Emission Control Areas (ECAs) and enabling access to environmentally sensitive maritime regions, including the Norwegian fjords.

A separate emissions evaluation was carried out on three river cruise vessels. Results showed a substantial reduction in particulate matter levels and average nitrogen oxide (NO_x) concentrations approximately 70% lower in the treated group compared to untreated vessels. Carbon monoxide (CO) emissions were also reduced, and hydrogen sulfide (H₂S) levels remained minimal across all trials. All tested vessels remained well within the regulatory thresholds for inland waterways. The emissions data provided valuable insight into how monitoring trends in unburned hydrocarbons can support targeted preventative maintenance scheduling and prioritization of engine inspections based on exhaust profiles.

In a third-party study onboard a superyacht, emissions testing was conducted across three engines with varying levels of treatment. Particulate emissions were reduced by a factor of approximately 16 to the cleanest partially treated engine. Black smoke emissions decreased by more than 90%, and when engine load increased to 36%, no significant change in particulate output was observed, suggesting stable combustion behavior under dynamic operating conditions.

Overall, these evaluations reinforce the role of the fuel conditioner in supporting emissions reduction and operational efficiency in the marine sector. They also serve as a foundation for broader onboard emissions monitoring programs planned for larger commercial vessels.

Other Land Evaluations

Several land-based evaluations have demonstrated the applicability of the fuel conditioner in reducing fuel consumption and emissions from support systems of the marine logistics and operations supply chain, particularly critical to port operations, inland freight and transport infrastructure.

A nine-month evaluation was conducted across two railway vehicle types, four tampers and six Class 66 diesel locomotives. Under high workload-conditions, the locomotives recorded an average fuel efficiency improvement of 4.5%. The tampers demonstrated a 4.1% reduction in fuel consumption measured in litres per mile, reflecting reduced fuel costs across long-haul operations, and 7.3% reduction in litres per hour, indicating maximized efficiency during idling and low-speed operations. Independent emissions testing across both vehicle types recorded substantial reductions. Carbon monoxide (CO) was reduced by up to 63.4%, sulfur oxides (SO_x) by 47.5%-96.8%, carbon dioxide (CO₂) by 9.1%-45.9%, nitrogen oxides (NO_x) by 4.3-47.3%, PM_{2.5} by 21.5%-32.6% and PM₁₀ by 2.5%-30.4%. It should be noted that particulate matter readings were obtained using a localized emissions sensor placed outside the exhaust system, which will under-report total particulate reductions compared to direct exhaust sampling.

Trials conducted on freight and logistics vehicles, such as construction and fuel transport trucks, recorded an average fuel consumption reduction of 8%. Evaluation on heavy-duty equipment, e.g. loaders and excavators, showed average reductions of 6%.

Conclusions

This paper concludes that the evaluated fuel conditioner offers a technically effective, commercially viable and operationally practical solution for delivering immediate reductions in fuel consumption and emissions across the marine logistics and operations supply chain. Its compatibility with all liquid hydrocarbon fuels and wide applicability across engine types enables deployment without the need for retrofitting, capital expenditure, or operational disruption.

The conditioner's formula has been verified to maintain ISO 8217 fuel compliance and has demonstrated enhanced lubricity performance. Independent evaluations confirmed consistent improvements in fuel efficiency, with savings ranging from 3% to 7% across marine vessels and rail applications, up to 15% in a controlled generator trial, and an average of 8% across land-based logistics. Emissions testing confirmed significant reductions in greenhouse gases and particulate matter (up to 96%). These outcomes were achieved under both real-world and controlled conditions, with additional benefits observed in engine cleanliness, improved injector condition, and lower exhaust gas temperatures, all contributing to smoother engine operation and reduced maintenance needs.

The conditioner also demonstrated full compatibility with blended biofuels (e.g., B30), with no adverse effects on performance. Further testing is planned on additional fuel types including LSMGO and higher bio-content fuels such as B100 and LSMGO, to assess the conditioner's impact on lubricity and long-term engine wear.

Overall, the findings presented in this paper demonstrate the broad applicability of the fuel conditioner across the entire marine logistics and operations supply chain. Its effective use in both vessel engines and land-based equipment supports not only global sustainability targets but also compliance with local air quality regulations. With the global potential to save up to 5.8 million barrels of oil every day (Statista 2024), the conditioner offers greater global reach than any other currently available direct fuel decarbonization solution. By enabling more complete and efficient combustion of existing fuels, it provides a credible, near-term pathway for vessel operators facing tightening environmental regulations. In doing so, it supports immediate progress toward emissions reductions, fuel efficiency, and operational resilience while the industry continues its transition toward long-term alternative fuel solutions.

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